pubhlth project

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R Markdown

Import datasets

```r
# SVI Dataset
county_raw = read_csv("../data/SVI_County_Export.csv", show_col_types = FALSE) |>
  clean_names()

# SVI dataset cleaning
county_svi = county_raw |>
  separate(location, into = c("county_name", "state"), sep = ", ") |>
  mutate(
    county = county_name |>
      str_remove(" County") |>
      str_to_upper()
  )
prep = function(df) {
  df |>
    mutate(across(where(is.character), as.factor)) |>
    mutate(across(where(is.factor), ~fct_explicit_na(.x, na_level = "Missing"))) |>
    mutate(across(where(is.numeric), ~ifelse(is.na(.x), median(.x, na.rm = TRUE), .x)))
}

county_svi_simple = prep(county_svi)
```

```
## Warning: There was 1 warning in ‘mutate()‘.
## i In argument: ‘across(where(is.factor), ~fct_explicit_na(.x, na_level =
##   "Missing"))‘.
## Caused by warning:
## ! ‘fct_explicit_na()‘ was deprecated in forcats 1.0.0.
## i Please use ‘fct_na_value_to_level()‘ instead.
```

```r
county_svi_simple$county_name <- gsub(" County$", "", county_svi_simple$county_name)
county_svi_simple$fips <- sprintf("%05d", county_svi_simple$fips)
head(county_svi_simple)
```

```
## # A tibble: 6 x 27
##   county_name state fips  x2018_overall_svi_score total_population housing_units
##   <chr>       <fct> <chr>                   <dbl>            <dbl>         <dbl>
```

```
## 1 Adams       Ohio  39001                    0.931            27878         12920
## 2 Allen       Ohio  39003                    0.770           103642         45063
## 3 Ashland     Ohio  39005                    0.575            53477         22250
## 4 Ashtabula   Ohio  39007                    1                98136         46107
## 5 Athens      Ohio  39009                    0.942            65936         26610
## 6 Auglaize    Ohio  39011                    0.023            45784         19852
## # i 21 more variables: occupied_households <dbl>, socioeconomic_score <dbl>,
## #   below_poverty <dbl>, unemployed <dbl>, income <dbl>, no_hs_diploma <dbl>,
## #   house_composition_disability_score <dbl>, age_65_or_older <dbl>,
## #   age_17_or_under <dbl>, civilian_with_a_disability <dbl>,
## #   single_parent_household <dbl>, minority_status_language <dbl>,
## #   minority <dbl>, speak_english_less_than_well <dbl>,
## #   housing_type_transportation <dbl>, multi_unit_structure <dbl>, ...
```

Create & Export Healthcare Access Dataset

```
# Healthcare access dataset
excel_sheets("../data/2025 County Health Rankings Ohio Data - v3.xlsx")
```

```
## [1] "Introduction"                   "Select Measure Data"
## [3] "Select Measure Sources & Years" "Additional Measure Data"
## [5] "Addtl Measure Sources & Years"  "Health Groups"
```

```
main <- read_excel("../data/2025 County Health Rankings Ohio Data - v3.xlsx",
                        sheet = "Select Measure Data", skip = 1)
```

```
## New names:
## * `Unreliable` -> `Unreliable...4`
## * `95% CI - Low` -> `95% CI - Low...7`
## * `95% CI - High` -> `95% CI - High...8`
## * `National Z-Score` -> `National Z-Score...9`
## * `95% CI - Low` -> `95% CI - Low...39`
## * `95% CI - High` -> `95% CI - High...40`
## * `National Z-Score` -> `National Z-Score...41`
## * `Unreliable` -> `Unreliable...42`
## * `95% CI - Low` -> `95% CI - Low...44`
## * `95% CI - High` -> `95% CI - High...45`
## * `National Z-Score` -> `National Z-Score...46`
## * `95% CI - Low` -> `95% CI - Low...69`
## * `95% CI - High` -> `95% CI - High...70`
## * `National Z-Score` -> `National Z-Score...71`
## * `95% CI - Low` -> `95% CI - Low...73`
## * `95% CI - High` -> `95% CI - High...74`
## * `National Z-Score` -> `National Z-Score...75`
## * `National Z-Score` -> `National Z-Score...77`
## * `National Z-Score` -> `National Z-Score...84`
## * `National Z-Score` -> `National Z-Score...86`
## * `National Z-Score` -> `National Z-Score...90`
## * `National Z-Score` -> `National Z-Score...94`
## * `National Z-Score` -> `National Z-Score...98`
## * `National Z-Score` -> `National Z-Score...100`
```

```
## * ‘National Z-Score‘ -> ‘National Z-Score...107‘
## * ‘95% CI - Low‘ -> ‘95% CI - Low...115‘
## * ‘95% CI - High‘ -> ‘95% CI - High...116‘
## * ‘National Z-Score‘ -> ‘National Z-Score...117‘
## * ‘95% CI - Low‘ -> ‘95% CI - Low...119‘
## * ‘95% CI - High‘ -> ‘95% CI - High...120‘
## * ‘National Z-Score‘ -> ‘National Z-Score...130‘
## * ‘95% CI - Low‘ -> ‘95% CI - Low...132‘
## * ‘95% CI - High‘ -> ‘95% CI - High...133‘
## * ‘National Z-Score‘ -> ‘National Z-Score...134‘
## * ‘95% CI - Low‘ -> ‘95% CI - Low...152‘
## * ‘95% CI - High‘ -> ‘95% CI - High...153‘
## * ‘National Z-Score‘ -> ‘National Z-Score...154‘
## * ‘National Z-Score‘ -> ‘National Z-Score...156‘
## * ‘National Z-Score‘ -> ‘National Z-Score...158‘
## * ‘95% CI - Low‘ -> ‘95% CI - Low...161‘
## * ‘95% CI - High‘ -> ‘95% CI - High...162‘
## * ‘National Z-Score‘ -> ‘National Z-Score...163‘
## * ‘National Z-Score‘ -> ‘National Z-Score...165‘
## * ‘Population‘ -> ‘Population...167‘
## * ‘95% CI - Low‘ -> ‘95% CI - Low...169‘
## * ‘95% CI - High‘ -> ‘95% CI - High...170‘
## * ‘National Z-Score‘ -> ‘National Z-Score...171‘
## * ‘Population‘ -> ‘Population...173‘
## * ‘95% CI - Low‘ -> ‘95% CI - Low...175‘
## * ‘95% CI - High‘ -> ‘95% CI - High...176‘
## * ‘National Z-Score‘ -> ‘National Z-Score...177‘
## * ‘National Z-Score‘ -> ‘National Z-Score...181‘
## * ‘National Z-Score‘ -> ‘National Z-Score...185‘
## * ‘95% CI - Low‘ -> ‘95% CI - Low...187‘
## * ‘95% CI - High‘ -> ‘95% CI - High...188‘
## * ‘National Z-Score‘ -> ‘National Z-Score...189‘
## * ‘95% CI - Low‘ -> ‘95% CI - Low...197‘
## * ‘95% CI - High‘ -> ‘95% CI - High...198‘
## * ‘National Z-Score‘ -> ‘National Z-Score...199‘
## * ‘National Z-Score‘ -> ‘National Z-Score...223‘
## * ‘National Z-Score‘ -> ‘National Z-Score...225‘
```

```r
healthcare_vars <- c(
  "FIPS", "State", "County",  # keep these for identification

  "# Primary Care Physicians",
  "Primary Care Physicians Rate",
  "Primary Care Physicians Ratio",
  "National Z-Score...90"
)

healthcare_access_data <- main[, healthcare_vars]
healthcare_access_data <- healthcare_access_data[-1, ]
head(healthcare_access_data)
```

```
## # A tibble: 6 x 7
##   FIPS  State County   ‘# Primary Care Physicians‘ Primary Care Physicians Ra~1
##   <chr> <chr> <chr>                          <dbl>                        <dbl>
```

```
## 1 39001 Ohio  Adams                               12                          43.6
## 2 39003 Ohio  Allen                               66                          64.9
## 3 39005 Ohio  Ashland                             23                          44.0
## 4 39007 Ohio  Ashtabula                           28                          28.8
## 5 39009 Ohio  Athens                              39                          62.8
## 6 39011 Ohio  Auglaize                            16                          34.7
## # i abbreviated name: 1: `Primary Care Physicians Rate`
## # i 2 more variables: `Primary Care Physicians Ratio` <chr>,
## #   `National Z-Score...90` <dbl>
```

Join into single dataset

```r
#inner join the two datasets by county, wsanity checked 54 colns
merged_data = healthcare_access_data %>%
  left_join(county_svi_simple, by = c("FIPS" = "fips"))

head(merged_data)
```

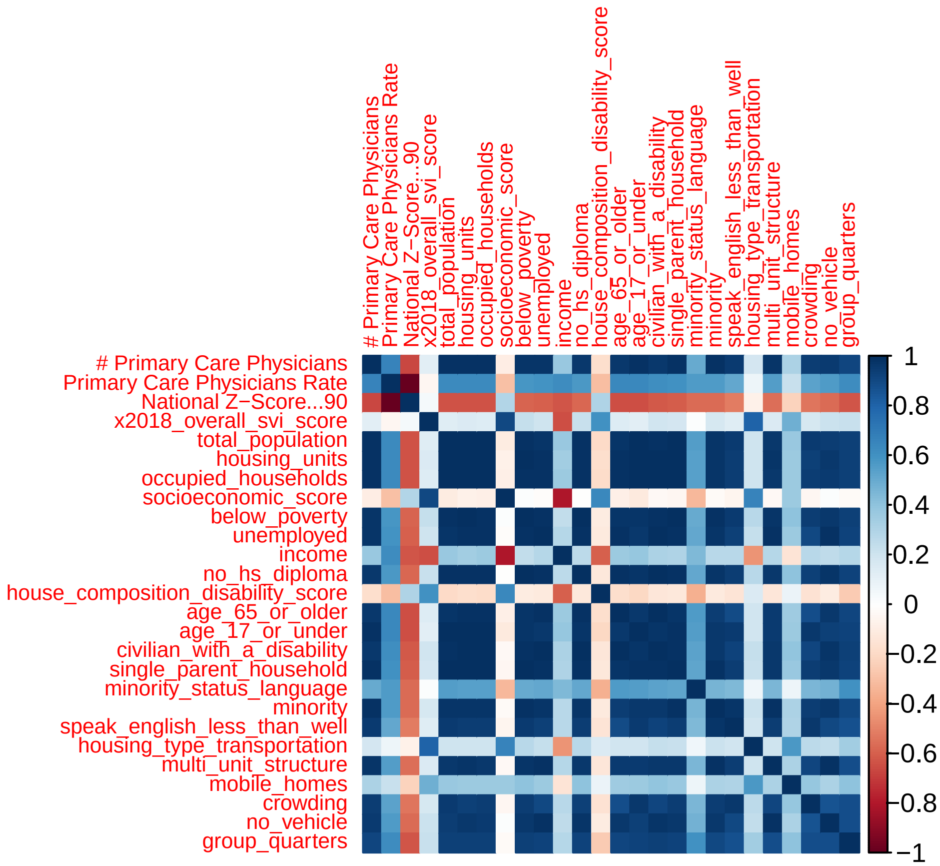
```
## # A tibble: 6 x 33
##   FIPS  State County    `# Primary Care Physicians` Primary Care Physicians Ra~1
##   <chr> <chr> <chr>                           <dbl>                        <dbl>
## 1 39001 Ohio  Adams                               12                          43.6
## 2 39003 Ohio  Allen                               66                          64.9
## 3 39005 Ohio  Ashland                             23                          44.0
## 4 39007 Ohio  Ashtabula                           28                          28.8
## 5 39009 Ohio  Athens                              39                          62.8
## 6 39011 Ohio  Auglaize                            16                          34.7
## # i abbreviated name: 1: `Primary Care Physicians Rate`
## # i 28 more variables: `Primary Care Physicians Ratio` <chr>,
## #   `National Z-Score...90` <dbl>, county_name <chr>, state <fct>,
## #   x2018_overall_svi_score <dbl>, total_population <dbl>, housing_units <dbl>,
## #   occupied_households <dbl>, socioeconomic_score <dbl>, below_poverty <dbl>,
## #   unemployed <dbl>, income <dbl>, no_hs_diploma <dbl>,
## #   house_composition_disability_score <dbl>, age_65_or_older <dbl>, ...
```

```r
print(colnames(merged_data))
```

```
##  [1] "FIPS"                               "State"
##  [3] "County"                             "# Primary Care Physicians"
##  [5] "Primary Care Physicians Rate"       "Primary Care Physicians Ratio"
##  [7] "National Z-Score...90"              "county_name"
##  [9] "state"                              "x2018_overall_svi_score"
## [11] "total_population"                   "housing_units"
## [13] "occupied_households"                "socioeconomic_score"
## [15] "below_poverty"                      "unemployed"
## [17] "income"                             "no_hs_diploma"
## [19] "house_composition_disability_score" "age_65_or_older"
## [21] "age_17_or_under"                    "civilian_with_a_disability"
## [23] "single_parent_household"            "minority_status_language"
## [25] "minority"                           "speak_english_less_than_well"
## [27] "housing_type_transportation"        "multi_unit_structure"
```

```
## [29] "mobile_homes"                          "crowding"
## [31] "no_vehicle"                            "group_quarters"
## [33] "county"
```

```r
# Correlation matrix for merged dataset
numeric_vars <- dplyr::select(merged_data, where(is.numeric))
cor_matrix = cor(numeric_vars, use = "pairwise.complete.obs")
corrplot(cor_matrix, method = "color", tl.cex = 0.7)
```



Fit model

```r
full_mod = lm(
  `Primary Care Physicians Rate` ~
    total_population +
    housing_units +
    occupied_households +
    socioeconomic_score +
    below_poverty +
    unemployed +
    income +
    no_hs_diploma +
    house_composition_disability_score +
    age_65_or_older +
```

```r
    age_17_or_under +
    civilian_with_a_disability +
    single_parent_household +
    minority_status_language +
    minority +
    speak_english_less_than_well +
    housing_type_transportation+
    multi_unit_structure +
    mobile_homes +
    crowding +
    no_vehicle +
    group_quarters,
  data = merged_data
)
summary(full_mod)
```

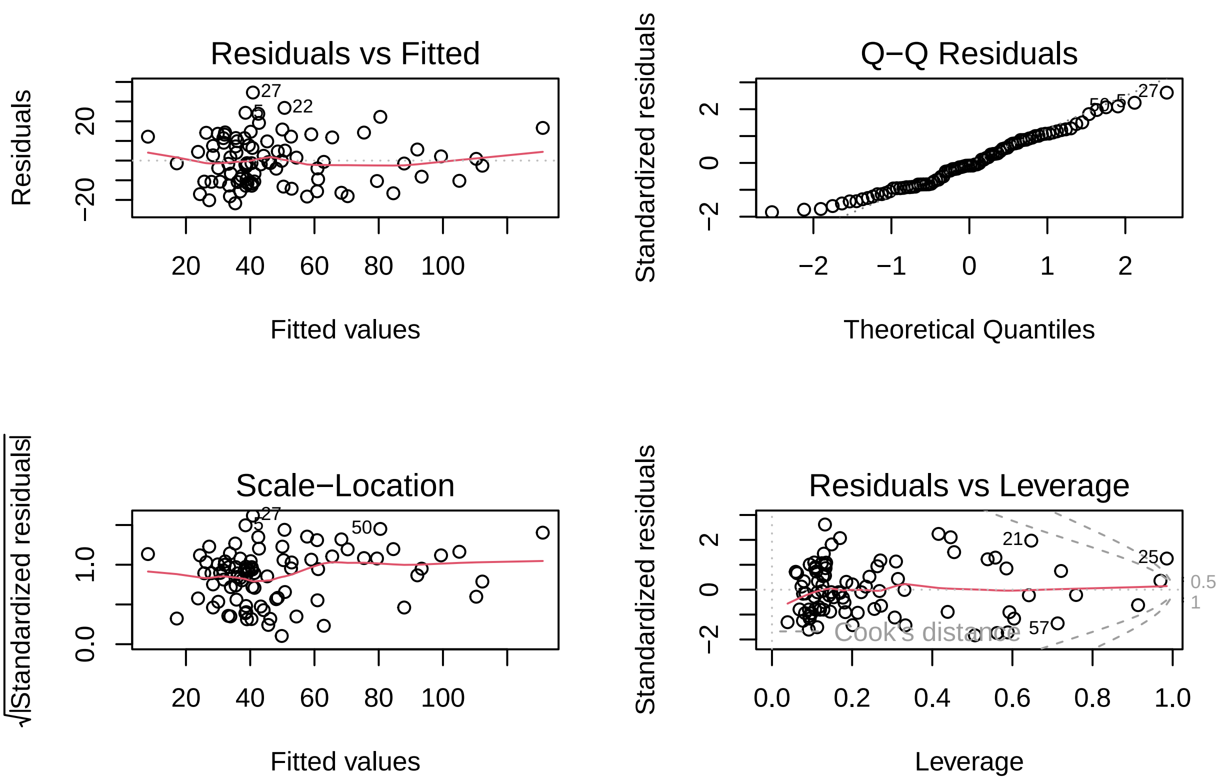
```
##
## Call:
## lm(formula = `Primary Care Physicians Rate` ~ total_population +
##     housing_units + occupied_households + socioeconomic_score +
##     below_poverty + unemployed + income + no_hs_diploma + house_composition_disability_score +
##     age_65_or_older + age_17_or_under + civilian_with_a_disability +
##     single_parent_household + minority_status_language + minority +
##     speak_english_less_than_well + housing_type_transportation +
##     multi_unit_structure + mobile_homes + crowding + no_vehicle +
##     group_quarters, data = merged_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -21.729 -10.641  -1.293   9.791  34.610
##
## Coefficients:
##                                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)                        -5.197e+01  2.909e+01  -1.786  0.07873 .
## total_population                   -2.217e-03  1.058e-03  -2.096  0.04001 *
## housing_units                      -2.032e-03  1.400e-03  -1.451  0.15148
## occupied_households                 5.681e-03  2.332e-03   2.436  0.01760 *
## socioeconomic_score                -1.588e+01  1.564e+01  -1.015  0.31374
## below_poverty                       3.547e-03  1.190e-03   2.981  0.00404 **
## unemployed                         -9.133e-04  4.953e-03  -0.184  0.85427
## income                              2.751e-03  9.569e-04   2.875  0.00546 **
## no_hs_diploma                      -4.229e-03  1.898e-03  -2.228  0.02938 *
## house_composition_disability_score  1.052e+01  1.082e+01   0.973  0.33436
## age_65_or_older                    -1.278e-04  1.763e-03  -0.072  0.94245
## age_17_or_under                     5.141e-03  2.186e-03   2.352  0.02172 *
## civilian_with_a_disability          1.517e-03  1.464e-03   1.036  0.30402
## single_parent_household            -1.317e-02  6.047e-03  -2.178  0.03303 *
## minority_status_language            8.571e+00  8.346e+00   1.027  0.30821
## minority                            2.430e-04  7.115e-04   0.342  0.73379
## speak_english_less_than_well        1.046e-03  6.749e-03   0.155  0.87731
## housing_type_transportation         3.392e+01  1.186e+01   2.860  0.00570 **
## multi_unit_structure               -1.365e-03  1.536e-03  -0.889  0.37731
## mobile_homes                        8.254e-04  2.803e-03   0.295  0.76931
```

```
## crowding                              -2.996e-02  1.153e-02  -2.598  0.01158 *
## no_vehicle                            -4.401e-04  3.117e-03  -0.141  0.88817
## group_quarters                         7.670e-04  1.615e-03   0.475  0.63640
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 14.2 on 65 degrees of freedom
## Multiple R-squared:  0.7736, Adjusted R-squared:  0.6969
## F-statistic: 10.09 on 22 and 65 DF,  p-value: 1.533e-13
```

```
# Visualizations

# Residuals plots
par(mfrow = c(2,2))
plot(full_mod)
```

```
## Warning in sqrt(crit * p * (1 - hh)/hh): NaNs produced
## Warning in sqrt(crit * p * (1 - hh)/hh): NaNs produced
```

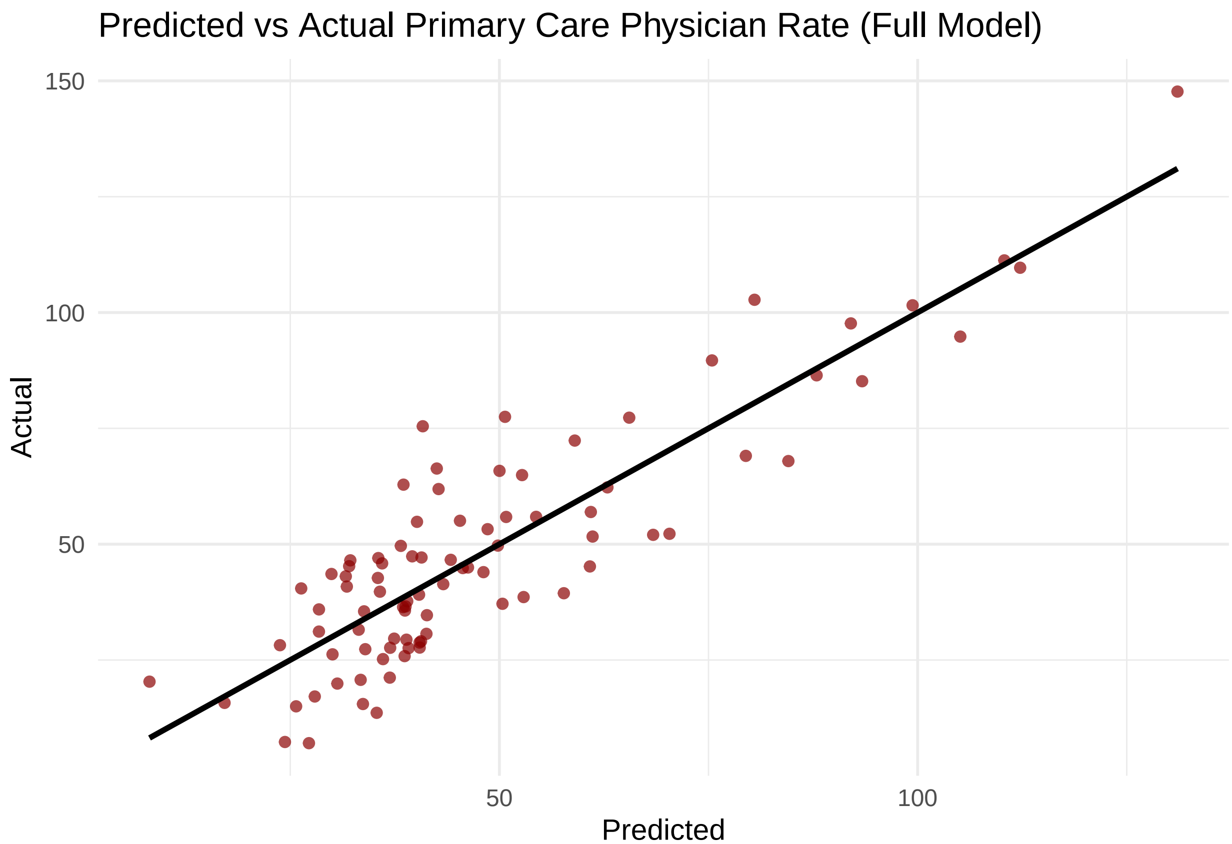



```
# PCP Prediction plot
merged_data$pred_full <- predict(full_mod)

ggplot(merged_data,
       aes(x = pred_full, y = `Primary Care Physicians Rate`)) +
  geom_point(alpha = 0.7, color = "darkred") +
```

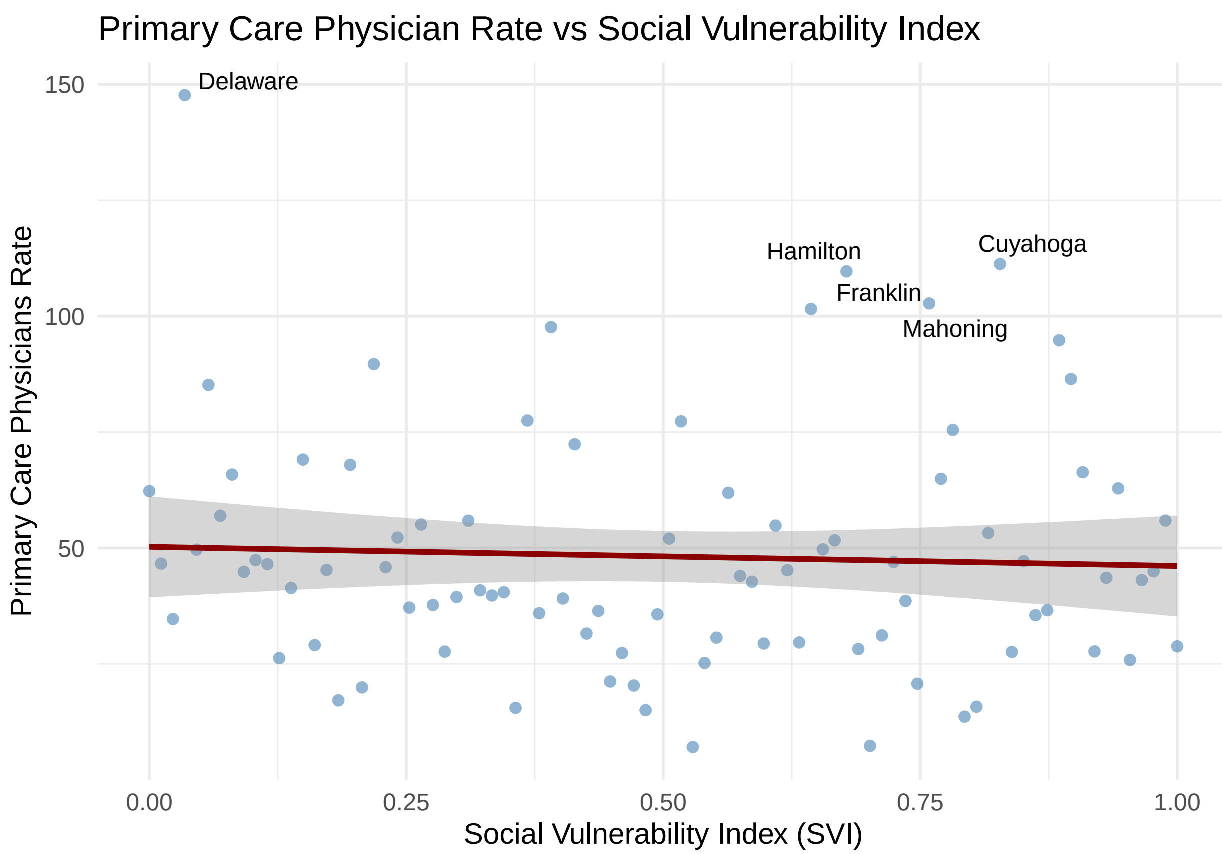
```r
  geom_smooth(method = "lm", color = "black", se = FALSE) +
  labs(
    title = "Predicted vs Actual Primary Care Physician Rate (Full Model)",
    x = "Predicted",
    y = "Actual"
  ) +
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




```r
# PCP vs SVI score
ggplot(merged_data, aes(x = x2018_overall_svi_score, y = `Primary Care Physicians Rate`)) +
  geom_point(alpha = 0.6, color = "steelblue") +
  geom_smooth(method = "lm", se = TRUE, color = "darkred") +
  geom_text_repel(data = subset(merged_data, `Primary Care Physicians Rate` > quantile(`Primary Care Phy
                  aes(label = county_name), size = 3, color = "black") +
  labs(
    title = "Primary Care Physician Rate vs Social Vulnerability Index",
    x = "Social Vulnerability Index (SVI)",
    y = "Primary Care Physicians Rate"
  ) +
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



Model Selection

```r
#vif(full_mod)
step_mod = stepAIC(full_mod, direction = "backward")
```

```
## Start:  AIC=486.36
## `Primary Care Physicians Rate` ~ total_population + housing_units +
##     occupied_households + socioeconomic_score + below_poverty +
##     unemployed + income + no_hs_diploma + house_composition_disability_score +
##     age_65_or_older + age_17_or_under + civilian_with_a_disability +
##     single_parent_household + minority_status_language + minority +
##     speak_english_less_than_well + housing_type_transportation +
##     multi_unit_structure + mobile_homes + crowding + no_vehicle +
##     group_quarters
##
##                                     Df Sum of Sq   RSS    AIC
## - age_65_or_older                    1      1.06 13115 484.37
## - no_vehicle                         1      4.02 13118 484.39
## - speak_english_less_than_well       1      4.85 13119 484.39
## - unemployed                         1      6.86 13121 484.41
## - mobile_homes                       1     17.50 13131 484.48
## - minority                           1     23.54 13137 484.52
## - group_quarters                     1     45.51 13159 484.66
## - multi_unit_structure               1    159.43 13273 485.42
```

```
## - house_composition_disability_score  1    190.85 13305 485.63
## - socioeconomic_score                 1    207.96 13322 485.74
## - minority_status_language            1    212.81 13327 485.78
## - civilian_with_a_disability          1    216.56 13330 485.80
## <none>                                            13114 486.36
## - housing_units                       1    425.01 13539 487.17
## - total_population                    1    886.03 14000 490.11
## - single_parent_household             1    957.17 14071 490.56
## - no_hs_diploma                       1   1001.10 14115 490.83
## - age_17_or_under                     1   1116.01 14230 491.55
## - occupied_households                 1   1197.28 14311 492.05
## - crowding                            1   1362.01 14476 493.06
## - housing_type_transportation         1   1649.74 14764 494.79
## - income                              1   1667.58 14782 494.89
## - below_poverty                       1   1793.16 14907 495.64
##
## Step:  AIC=484.37
## `Primary Care Physicians Rate` ~ total_population + housing_units +
##     occupied_households + socioeconomic_score + below_poverty +
##     unemployed + income + no_hs_diploma + house_composition_disability_score +
##     age_17_or_under + civilian_with_a_disability + single_parent_household +
##     minority_status_language + minority + speak_english_less_than_well +
##     housing_type_transportation + multi_unit_structure + mobile_homes +
##     crowding + no_vehicle + group_quarters
##
##                                      Df Sum of Sq   RSS    AIC
## - no_vehicle                          1      5.14 13120 482.40
## - speak_english_less_than_well        1      5.64 13121 482.40
## - unemployed                          1      7.94 13123 482.42
## - mobile_homes                        1     26.32 13141 482.54
## - minority                            1     27.26 13142 482.55
## - group_quarters                      1     46.26 13161 482.68
## - multi_unit_structure                1    167.95 13283 483.49
## - house_composition_disability_score  1    196.28 13311 483.67
## - socioeconomic_score                 1    207.74 13323 483.75
## - minority_status_language            1    211.75 13327 483.78
## - civilian_with_a_disability          1    228.68 13344 483.89
## <none>                                            13115 484.37
## - housing_units                       1    498.20 13613 485.65
## - total_population                    1    886.34 14001 488.12
## - single_parent_household             1   1052.40 14167 489.16
## - age_17_or_under                     1   1194.86 14310 490.04
## - occupied_households                 1   1243.88 14359 490.34
## - no_hs_diploma                       1   1250.09 14365 490.38
## - crowding                            1   1411.46 14526 491.36
## - housing_type_transportation         1   1648.71 14764 492.79
## - income                              1   1724.32 14839 493.24
## - below_poverty                       1   1815.14 14930 493.77
##
## Step:  AIC=482.4
## `Primary Care Physicians Rate` ~ total_population + housing_units +
##     occupied_households + socioeconomic_score + below_poverty +
##     unemployed + income + no_hs_diploma + house_composition_disability_score +
##     age_17_or_under + civilian_with_a_disability + single_parent_household +
```

```
##     minority_status_language + minority + speak_english_less_than_well +
##     housing_type_transportation + multi_unit_structure + mobile_homes +
##     crowding + group_quarters
##
##                                   Df Sum of Sq   RSS    AIC
## - unemployed                       1     13.26 13133 480.49
## - speak_english_less_than_well     1     16.28 13136 480.51
## - minority                         1     22.51 13143 480.55
## - mobile_homes                     1     38.13 13158 480.66
## - group_quarters                   1     48.85 13169 480.73
## - house_composition_disability_score  1    194.37 13314 481.70
## - multi_unit_structure             1    195.77 13316 481.70
## - socioeconomic_score              1    202.60 13323 481.75
## - minority_status_language         1    207.62 13328 481.78
## - civilian_with_a_disability       1    236.57 13357 481.97
## <none>                                         13120 482.40
## - housing_units                    1    517.57 13638 483.81
## - total_population                 1    971.99 14092 486.69
## - age_17_or_under                  1   1207.54 14328 488.15
## - occupied_households              1   1351.19 14471 489.03
## - single_parent_household          1   1355.59 14476 489.05
## - crowding                         1   1438.36 14558 489.56
## - housing_type_transportation      1   1754.31 14874 491.45
## - no_hs_diploma                    1   1785.87 14906 491.63
## - income                           1   1786.68 14907 491.64
## - below_poverty                    1   1828.20 14948 491.88
##
## Step:  AIC=480.49
## `Primary Care Physicians Rate` ~ total_population + housing_units +
##     occupied_households + socioeconomic_score + below_poverty +
##     income + no_hs_diploma + house_composition_disability_score +
##     age_17_or_under + civilian_with_a_disability + single_parent_household +
##     minority_status_language + minority + speak_english_less_than_well +
##     housing_type_transportation + multi_unit_structure + mobile_homes +
##     crowding + group_quarters
##
##                                   Df Sum of Sq   RSS    AIC
## - speak_english_less_than_well     1     16.27 13150 478.60
## - minority                         1     19.22 13153 478.62
## - group_quarters                   1     50.29 13184 478.83
## - mobile_homes                     1     53.39 13187 478.85
## - house_composition_disability_score  1    189.64 13323 479.75
## - socioeconomic_score              1    198.82 13332 479.81
## - minority_status_language         1    215.65 13349 479.92
## - civilian_with_a_disability       1    226.53 13360 480.00
## - multi_unit_structure             1    237.28 13371 480.07
## <none>                                         13133 480.49
## - housing_units                    1    561.30 13695 482.17
## - total_population                 1   1041.75 14175 485.21
## - age_17_or_under                  1   1279.36 14413 486.67
## - single_parent_household          1   1376.95 14510 487.26
## - occupied_households              1   1471.66 14605 487.84
## - crowding                         1   1671.57 14805 489.03
## - housing_type_transportation      1   1752.21 14886 489.51
```

```
## - income                            1   1778.39 14912 489.67
## - no_hs_diploma                     1   1832.88 14966 489.99
## - below_poverty                     1   1878.57 15012 490.26
##
## Step:  AIC=478.6
## `Primary Care Physicians Rate` ~ total_population + housing_units +
##     occupied_households + socioeconomic_score + below_poverty +
##     income + no_hs_diploma + house_composition_disability_score +
##     age_17_or_under + civilian_with_a_disability + single_parent_household +
##     minority_status_language + minority + housing_type_transportation +
##     multi_unit_structure + mobile_homes + crowding + group_quarters
##
##                                    Df Sum of Sq   RSS    AIC
## - group_quarters                    1     34.05 13184 476.83
## - mobile_homes                      1     63.96 13214 477.03
## - house_composition_disability_score  1    200.88 13350 477.93
## - civilian_with_a_disability        1    210.51 13360 478.00
## - socioeconomic_score               1    224.39 13374 478.09
## - minority_status_language          1    238.39 13388 478.18
## - minority                          1    264.02 13414 478.35
## <none>                                          13150 478.60
## - multi_unit_structure              1    609.66 13759 480.59
## - housing_units                     1    617.99 13768 480.64
## - occupied_households               1   1463.65 14613 485.89
## - single_parent_household           1   1579.51 14729 486.58
## - total_population                  1   1646.51 14796 486.98
## - housing_type_transportation       1   1744.38 14894 487.56
## - crowding                          1   1828.80 14978 488.06
## - income                            1   1830.17 14980 488.07
## - below_poverty                     1   1899.03 15049 488.47
## - no_hs_diploma                     1   2004.04 15154 489.08
## - age_17_or_under                   1   2211.05 15361 490.28
##
## Step:  AIC=476.83
## `Primary Care Physicians Rate` ~ total_population + housing_units +
##     occupied_households + socioeconomic_score + below_poverty +
##     income + no_hs_diploma + house_composition_disability_score +
##     age_17_or_under + civilian_with_a_disability + single_parent_household +
##     minority_status_language + minority + housing_type_transportation +
##     multi_unit_structure + mobile_homes + crowding
##
##                                    Df Sum of Sq   RSS    AIC
## - mobile_homes                      1     47.34 13231 475.14
## - house_composition_disability_score  1    168.10 13352 475.94
## - socioeconomic_score               1    216.57 13400 476.26
## - civilian_with_a_disability        1    241.93 13426 476.43
## - minority                          1    268.26 13452 476.60
## - minority_status_language          1    276.68 13460 476.65
## <none>                                          13184 476.83
## - housing_units                     1    599.79 13784 478.74
## - multi_unit_structure              1    621.04 13805 478.88
## - occupied_households               1   1465.55 14649 484.10
## - single_parent_household           1   1545.79 14730 484.58
## - total_population                  1   1739.23 14923 485.73
```

```
## - income                            1   1821.85 15006 486.22
## - crowding                          1   1866.16 15050 486.48
## - below_poverty                     1   1978.11 15162 487.13
## - housing_type_transportation       1   2222.76 15406 488.54
## - age_17_or_under                   1   2226.88 15410 488.56
## - no_hs_diploma                     1   2338.85 15522 489.20
##
## Step:  AIC=475.14
## `Primary Care Physicians Rate` ~ total_population + housing_units +
##     occupied_households + socioeconomic_score + below_poverty +
##     income + no_hs_diploma + house_composition_disability_score +
##     age_17_or_under + civilian_with_a_disability + single_parent_household +
##     minority_status_language + minority + housing_type_transportation +
##     multi_unit_structure + crowding
##
##                                     Df Sum of Sq   RSS    AIC
## - house_composition_disability_score  1    216.86 13448 474.57
## - socioeconomic_score               1    218.86 13450 474.59
## - minority                          1    231.02 13462 474.67
## - minority_status_language          1    234.99 13466 474.69
## - civilian_with_a_disability        1    250.84 13482 474.80
## <none>                                          13231 475.14
## - housing_units                     1    562.07 13793 476.80
## - multi_unit_structure              1    575.18 13806 476.89
## - occupied_households               1   1438.03 14669 482.22
## - single_parent_household           1   1623.08 14854 483.33
## - total_population                  1   1700.20 14931 483.78
## - crowding                          1   1819.74 15051 484.48
## - income                            1   1959.24 15190 485.29
## - age_17_or_under                   1   2199.96 15431 486.68
## - no_hs_diploma                     1   2381.09 15612 487.70
## - housing_type_transportation       1   2522.31 15753 488.50
## - below_poverty                     1   2649.10 15880 489.20
##
## Step:  AIC=474.57
## `Primary Care Physicians Rate` ~ total_population + housing_units +
##     occupied_households + socioeconomic_score + below_poverty +
##     income + no_hs_diploma + age_17_or_under + civilian_with_a_disability +
##     single_parent_household + minority_status_language + minority +
##     housing_type_transportation + multi_unit_structure + crowding
##
##                               Df Sum of Sq   RSS    AIC
## - socioeconomic_score          1     66.27 13514 473.01
## - minority_status_language     1    239.16 13687 474.12
## - minority                     1    287.07 13735 474.43
## <none>                                    13448 474.57
## - civilian_with_a_disability   1    431.02 13879 475.35
## - housing_units                1    589.13 14037 476.35
## - multi_unit_structure         1    689.81 14138 476.98
## - single_parent_household      1   1476.53 14924 481.74
## - occupied_households          1   1566.88 15015 482.27
## - crowding                     1   1629.72 15078 482.64
## - income                       1   1941.06 15389 484.44
## - total_population             1   2077.53 15525 485.21
```

```
## - housing_type_transportation  1   2348.57 15796 486.74
## - age_17_or_under              1   2387.63 15836 486.96
## - below_poverty                1   2446.83 15895 487.28
## - no_hs_diploma                1   2543.88 15992 487.82
##
## Step:  AIC=473.01
## `Primary Care Physicians Rate` ~ total_population + housing_units +
##     occupied_households + below_poverty + income + no_hs_diploma +
##     age_17_or_under + civilian_with_a_disability + single_parent_household +
##     minority_status_language + minority + housing_type_transportation +
##     multi_unit_structure + crowding
##
##                              Df Sum of Sq   RSS    AIC
## - minority                    1     243.5 13758 472.58
## <none>                                    13514 473.01
## - minority_status_language    1     330.1 13844 473.13
## - civilian_with_a_disability  1     389.4 13904 473.51
## - housing_units               1     553.6 14068 474.54
## - multi_unit_structure        1     635.2 14149 475.05
## - single_parent_household     1    1411.4 14926 479.75
## - occupied_households         1    1502.4 15016 480.28
## - crowding                    1    1597.6 15112 480.84
## - total_population            1    2028.4 15542 483.31
## - below_poverty               1    2390.5 15905 485.34
## - age_17_or_under             1    2417.7 15932 485.49
## - no_hs_diploma               1    2479.1 15993 485.83
## - housing_type_transportation 1    2858.6 16373 487.89
## - income                      1    3920.5 17435 493.42
##
## Step:  AIC=472.58
## `Primary Care Physicians Rate` ~ total_population + housing_units +
##     occupied_households + below_poverty + income + no_hs_diploma +
##     age_17_or_under + civilian_with_a_disability + single_parent_household +
##     minority_status_language + housing_type_transportation +
##     multi_unit_structure + crowding
##
##                              Df Sum of Sq   RSS    AIC
## - civilian_with_a_disability  1     249.2 14007 472.16
## - minority_status_language    1     292.4 14050 472.43
## <none>                                    13758 472.58
## - housing_units               1     419.8 14177 473.22
## - multi_unit_structure        1     420.6 14178 473.23
## - single_parent_household     1    1179.2 14937 477.81
## - occupied_households         1    1262.8 15020 478.31
## - crowding                    1    1465.0 15223 479.48
## - total_population            1    1797.2 15555 481.38
## - age_17_or_under             1    2242.0 16000 483.86
## - no_hs_diploma               1    2313.0 16071 484.25
## - housing_type_transportation 1    2622.6 16380 485.93
## - below_poverty               1    2735.0 16493 486.53
## - income                      1    4121.3 17879 493.64
##
## Step:  AIC=472.16
## `Primary Care Physicians Rate` ~ total_population + housing_units +
```

```
##     occupied_households + below_poverty + income + no_hs_diploma +
##     age_17_or_under + single_parent_household + minority_status_language +
##     housing_type_transportation + multi_unit_structure + crowding
##
##                               Df Sum of Sq   RSS    AIC
## - minority_status_language     1     205.4 14212 471.44
## - housing_units                1     285.6 14292 471.93
## <none>                                     14007 472.16
## - multi_unit_structure         1     619.2 14626 473.96
## - single_parent_household      1     944.0 14951 475.90
## - occupied_households          1    1045.5 15052 476.49
## - total_population             1    1552.4 15559 479.41
## - age_17_or_under              1    1999.1 16006 481.90
## - no_hs_diploma                1    2307.4 16314 483.58
## - below_poverty                1    2738.8 16746 485.87
## - housing_type_transportation  1    2911.5 16918 486.78
## - crowding                     1    3009.7 17016 487.29
## - income                       1    4367.2 18374 494.04
##
## Step:  AIC=471.44
## `Primary Care Physicians Rate` ~ total_population + housing_units +
##     occupied_households + below_poverty + income + no_hs_diploma +
##     age_17_or_under + single_parent_household + housing_type_transportation +
##     multi_unit_structure + crowding
##
##                               Df Sum of Sq   RSS    AIC
## <none>                                     14212 471.44
## - housing_units                1     353.5 14566 471.60
## - multi_unit_structure         1     679.0 14891 473.54
## - single_parent_household      1     807.8 15020 474.30
## - occupied_households          1    1035.2 15247 475.62
## - total_population             1    1394.9 15607 477.68
## - age_17_or_under              1    1848.1 16060 480.20
## - no_hs_diploma                1    2233.3 16446 482.28
## - below_poverty                1    2680.3 16892 484.64
## - housing_type_transportation  1    3258.0 17470 487.60
## - crowding                     1    3414.6 17627 488.39
## - income                       1    5192.3 19404 496.84
```

```r
AIC(full_mod, step_mod)
```

```
##          df      AIC
## full_mod 24 738.0931
## step_mod 13 723.1710
```

```r
BIC(full_mod, step_mod)
```

```
##          df      BIC
## full_mod 24 797.5492
## step_mod 13 755.3764
```

```r
summary(full_mod)$adj.r.squared
```

```
## [1] 0.6969244
```

```r
summary(step_mod)$adj.r.squared
```

```
## [1] 0.7190809
```

```r
names(step_mod$coefficients)[-1]
```

```
##  [1] "total_population"            "housing_units"
##  [3] "occupied_households"         "below_poverty"
##  [5] "income"                      "no_hs_diploma"
##  [7] "age_17_or_under"             "single_parent_household"
##  [9] "housing_type_transportation" "multi_unit_structure"
## [11] "crowding"
```

```r
summary(step_mod)
```

```
##
## Call:
## lm(formula = `Primary Care Physicians Rate` ~ total_population +
##     housing_units + occupied_households + below_poverty + income +
##     no_hs_diploma + age_17_or_under + single_parent_household +
##     housing_type_transportation + multi_unit_structure + crowding,
##     data = merged_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -24.207 -10.276  -1.199   9.152  34.792
##
## Coefficients:
##                               Estimate Std. Error t value Pr(>|t|)
## (Intercept)                 -6.314e+01  1.743e+01  -3.622 0.000525 ***
## total_population            -1.332e-03  4.877e-04  -2.731 0.007842 **
## housing_units               -1.526e-03  1.110e-03  -1.375 0.173206
## occupied_households          3.714e-03  1.579e-03   2.353 0.021220 *
## below_poverty                3.423e-03  9.042e-04   3.786 0.000304 ***
## income                       3.349e-03  6.355e-04   5.269 1.24e-06 ***
## no_hs_diploma               -3.395e-03  9.823e-04  -3.456 0.000901 ***
## age_17_or_under              3.523e-03  1.121e-03   3.144 0.002380 **
## single_parent_household     -7.543e-03  3.629e-03  -2.078 0.041050 *
## housing_type_transportation  2.864e+01  6.861e+00   4.174 7.89e-05 ***
## multi_unit_structure        -1.071e-03  5.623e-04  -1.906 0.060491 .
## crowding                    -2.821e-02  6.601e-03  -4.273 5.52e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 13.67 on 76 degrees of freedom
## Multiple R-squared:  0.7546, Adjusted R-squared:  0.7191
## F-statistic: 21.25 on 11 and 76 DF,  p-value: < 2.2e-16
```